NARAYANA IIT-JEE ACADEMY-INDIA

SR STAR CO SC(MODEL-A&B) **Date: 25.07.2024**

Duration: 1 Hr **Max. Marks: 200**

CHEMISTRY NCERT TEST – 4

General Instructions:

- **Attempt all the questions, each correct answer carries +4 and wrong answer -1.**

SYLLABUS: 17th & 18th GROUP ELEMENTS

1. The F^- ions make the enamel on teeth much harder by converting hydroxyapatite (the enamel on the surface of teeth) into much harder fluoroapatite having the formula.

1) $[3\left(Ca_3\left(PO_4\right)_2\right).CaF_2]$

2) $[3\left(Ca_2\left(PO_4\right)_2\right).Ca\left(OH\right)_2]$

3) $[3\left(Ca_3\left(PO_4\right)_3\right)CaF_2]$

4) $[3\left(Ca_3\left(PO_4\right)_2\right)Ca\left(OH\right)_2]$

2. Element not showing variable oxidation state is

1) Bromine 2) Iodine 3) Chlorine 4) Fluorine

3. Statement - I: Fluorine forms one oxoacid.

Statement - II: Fluorine has smallest size amongst all halogens and is highly electronegative.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Statement I false, statement II is true

2) Both statement I & statement II are true

3) Both statement I & statement II are false

4) Statement I true, statement II is false

4. HF has highest boiling point among hydrogen halides because it has

1) strongest van der Waal's interactions

2) lowest ionic character

3) strongest hydrogen bonding

4) lowest dissociation enthalpy

5. Aqueous solution of which salt will not contain any ions with the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6$?

1) NaF 2) KBr 3) NaCl 4) CaI_2

6. The correct order of the thermal stability of hydrogen halides (H – X) is

1) HI > HCl < HF > HBr
2) HCl < HF > HBr < HI
3) HF > HCl > HBr > HI
4) HI < HBr > HCl < HF

7. Which among the following factors is the most important in making fluorine, the strongest oxidizing halogen?

(A) Hydration enthalpy
(B) Ionization enthalpy
(C) Electron affinity
(D) Bond dissociation energy

Correct one is

1) A & B only
2) B & C only
3) B & D only
4) A & D only

8. Assertion (A): Noble gases have very low melting and boiling points.

Reason (R): The atoms in these elements held together by weak dispersion forces.

In the light of the above statements. Choose the correct answer from the options given below

1) Both A and R are correct and R is the correct explanation of A
2) Both A and R are correct but R is NOT the correct explanation of A
3) A is not correct but R is correct
4) A is correct but R is not correct

9. Inert gases have positive electron gain enthalpy values. Its correct order is

1) Xe < Kr < Ne < He
2) He < Ne < Kr < Xe
3) He < Xe < Kr < Ne
4) He < Kr < Xe < Ne

10. Assertion A: Helium is used as a diluent for oxygen in modern diving apparatus.

Reason R: Helium has very low solubility in blood.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both A and R are correct and R is the correct explanation of A
2) Both A and R are correct but R is NOT the correct explanation of A
3) A is not correct but R is correct
4) A is correct but R is not correct

11. Statement-I: Ionization potential of xenon is nearly same as that of ionization potential of atomic oxygen.

Statement-II: $Xe[PtF_6]$ is a red coloured compound.

In the light of the above statements, choose the correct answer from the options given below:

1) Both Statement I and Statement II are true
2) Statement I is false but Statement II is true
3) Statement I is true but Statement II is false
4) Both Statement I and Statement II are false

12. Regarding XeF_4 , the incorrect statement is

1) Orientation of electron pairs in its structure is octahedral

2) Hydrolysis is a disproportionation reaction

3) It acts like fluoride ion acceptor with alkali metal fluorides

4) Hybridisation of Xe in XeF_4 is sp^3

13. Identify the incorrect statements.

(A) XeF_2 , XeF_4 , XeF_6 are colourless crystalline solids

(B) XeF_2 , XeF_4 , XeF_6 sublime at 298 K

(C) XeO_3 is a colourless explosive liquid

(D) $XeOF_4$ is a colourless volatile solid

1) A, C and D only 2) A, B and C only 3) C and D only 4) A, B, C and D

14. Match List-I with List-II:

List-I (Trend)	List-II (Property)
A. Xe > Kr >Ar> Ne > He	I. Abundance in atmosphere
B. Ne >Ar = Kr > Xe > He	II. Ionization potential
C. Ar> Ne > He > Kr > Xe	III. Positive electron gain enthalpy
D. He > Ne >Ar> Kr > Xe	IV. Melting point

Choose the correct answer from the options given below:

1) A-IV, B-I, C-II, D-IV 2) A-IV, B-III, C-I, D-II

3) A-II, B-I, C-IV, D-III 4) A-III, B-I, C-II, D-IV

15. $Xe(g)+F_2(g)\xrightarrow[60-70bar]{573K}A_{(s)}\xrightarrow{2H_2O}B+4HF$

$(1 \quad : \quad 20)$

Hybridization of xenon in the compounds 'A' and 'B' are respectively

1) sp^3d^3, sp^3d 2) sp^3d^3, sp^3d^2 3) sp^3d^3, sp^3 4) sp^3d^2, sp^3d^2

16. On addition of conc. H_2SO_4 to a chloride salt, colourless fumes are evolved but in case of iodide salt, violet fumes come out. This is because

1) H_2SO_4 reduces HI to I_2 2) HI is of violet colour

3) HI gets oxidised to I_2 4) HI changes to HIO_3

17. A black compound of manganese reacts with a halogen acid to give greenish yellow gas. When excess of this gas reacts with NH_3 an unstable trihalide is formed. In this process the oxidation state of nitrogen changes from __________

1) – 3 to +3 2) – 3 to 0 3) – 3 to +5 4) 0 to – 3

18. When chlorine gas is passed through hot and conc. NaOH solution, two changes are observed with respect to the oxidation number of chlorine during the reaction. These are _________ and __________.

(A) 0 to +5 (B) 0 to -5 (C) 0 to – 1 (D) 0 to +1

1) A & B only 2) C & D only 3) A & C only 4) A, B, C & D

19. Which of the following options are not in accordance with the property mentioned against them?

(i) $F_2 > Cl_2 > Br_2 > I_2$ Oxidising power.

(ii) MI > MBr > MCl > MF Ionic character of metal halide.

(iii) $F_2 > Cl_2 > Br_2 > I_2$ Bond dissociation enthalpy.

(iv) HI < HBr < HCl < HF Hydrogen-halogen bond strength.

1) (i) & (iii) only
2) (ii) & (iii) only
3) (iii) & (iv) only
4) (i), (ii), (iii) & (iv)

20. **Statement I**: The iodine oxides, I_2O_4 , I_2O_5 and I_2O_7 are soluble solids and decompose on heating.

Statement II: I_2O_5 is a good oxidising agent used in the estimation of carbon monoxide.

In the light of the above statements, choose the correct answer from the options given below:

1) Both Statement I and Statement II are true
2) Statement I is false but Statement II is true
3) Statement I is true but Statement II is false
4) Both Statement I and Statement II are false

21. Assertion (A): ICl is more reactive than I_2.

Reason (R): I – Cl bond is weaker than I – I bond.

In the light of the above statements choose the most appropriate answer from the options given below:

1) Both A and R are correct and R is the correct explanation of A
2) Both A and R are correct but R is NOT the correct explanation of A
3) A is not correct but R is correct
4) A is correct but R is not correct

22. Match the interhalogen compounds column-I with the geometry in column-II and assign the correct code.

Column I	Column II
(A) XX'	(i) T-shape
(B) XX'_3	(ii) Petagonal bipyramidal
(C) XX'_5	(iii) Linear
(D) XX'_7	(iv) Square pyramidal

1) A-(iii), B-(i), C-(iv), D(ii)
2) A-(v), B-(iv), C-(iii), D(ii)
3) A-(iv), B-(iii), C-(ii), D(i)
4) A-(iii), B-(iv), C-(i), D(ii)

23. Assertion (A): Helium has the lowest boiling point of any known substance.

Reason (R): Neon has an unusual property of diffusing through most commonly used laboratory materials such as rubber, glass or plastics.

In the light of the above statements, choose the correct answer from the options given below”

1) A is true, but R is false

2) A is false but R is true

3) Both A and R are true and R is the correct explanation of A

4) Both A and R are true and R is not the correct explanation of A

24. Noble gases are least reactive. Their inertness to chemical reactivity is attributed to the reason/s that

1) Their large positive values of electron gain enthalpy.

2) The noble gases except helium ($1s^2$) have completely filled ns^2np^6 electronic configuration in their valence shell.

3) They have high ionisation enthalpy.

4) All of these.

25. Which of the following statements are correct regarding inter halogen compounds?

(i) Among halogens, radius ratio between iodine and fluorine is maximum.

(ii) X–X′ bond in inter halogens is weaker than X–X bond in halogens except F–F bond.

(iii) Among interhalogen compounds maximum number of atoms are present in chlorine fluoride.

(iv) Interhalogen compounds are more reactive than halogen compounds

1) (i), (ii), (iii) only　　2) (ii), (iii), (iv) only

3) (i), (ii), (iv) only　　4) (i), (ii), (iii) & (iv)

26. In the preparation of compounds of Xe, Bartlett had taken $O_2^+PtF_6^-$ as a base compound. This is because

1) both O_2 and Xe have same size.

2) both O_2 and Xe have same electron gain enthalpy

3) both O_2 and Xe have almost same ionisation enthalpy.

4) both Xe and O_2 are gases.

27. Which of the following statements are true?

(i) The main commercial source of helium is natural gas.

(ii) Radon is obtained as a decay product of $^{226}_{88}Ra$

(iii) Hydrolysis of XeF_2 is a disproportionation reaction.

(iv) Their atmospheric abundance in dry air is ~ 1% by mass of which argon is the major constituent.

1) (i), (ii), (iii) only 2) (ii), (iii), (iv) only 3) (i) & (ii) only 4) (i), (ii), (iii) & (iv)

28. Match the compounds given in Column I with the hybridisation and shape given in Column II and mark the correct option.

Column I	**Column II**
(A) XeF_6	(1) sp^3d^3-distorted octahedral
(B) XeO_3	(2) sp^3d^2- square planar
(C) $XeOF_4$	(3) sp^3-pyramidal
(D) XeF_4	(4) sp^3d^2 square pyramidal

1) (A)-1, (B)-3, (C)-4, (D)-2 2) (A)-1, (B)-2, (C)-4, (D)-3
3) (A)-4, (B)-3, (C)-1, (D)-2 4) (A)-4, (B)-1, (C)-2, (D)-3

29. Match the items of Columns I and II and mark the correct option.

Column I	Column II
(A) Its partial hydrolysis does not change oxidation state of central atom	(1) He
(B) It is used as cryogenic agent.	(2) Ne
(C) It is used to provide inert atmosphere for filling electrical bulbs	(3) XeF_6
(D) It is used in bulbs that are used in botanical gardens and in green houses.	(4) Ar

1) A (1) B (4) C (2) D (3) 2) A (1) B (2) C (3) D (4)
3) A (3) B (1) C (4) D (2) 4) A (1) B (3) C (2) D (4)

30. XeF_4 upon reaction with SbF_5 gives the ion?

1) $[XeF_6]^{2-}$ 2) $[XeF_2]^{2+}$ 3) $[XeF_5]^{-}$ 4) $[XeF_3]^{+}$

31. Which of the following is sparingly soluble in water

1) Cl_2 2) Br_2 3) I_2 4) NaCl

32. Oxidising strength of oxyacids of chlorine is

1) $HClO_4 > HClO_3 > HClO_2 > HClO$
2) $HClO_4 > HClO_2 > HClO_3 > HClO$
3) $HClO > HClO_2 > HClO_3 > HClO_4$
4) $HClO > HClO_3 > HClO_2 > HClO_4$

33. Sea weeds are important source of

1) iron 2) chlorine 3) iodine 4) bromine

34. In the preparation of chlorine from HCl, MnO_2 acts as

1) oxidising agent 2) reducing agent
3) catalytic agent 4) dehydrating agent

35. Bromine (molecular mass 160 amu) boils at 59°C, whereas iodine monochloride (molecular mass 162 amu) boils at 92°C. The best explanation for this difference in their boiling point is

1) the difference in molecular masses

2) the I-Cl bond is stronger than Br-Br bond

3) Br_2 is a non-polar molecule, ICl is polar

4) The vander waals forces in ICl are

36. Which of the following species are present in chlorine water

1) only HCl 2) only HClO 3) HCl and $HClO_2$ 4) HCl and HClO

37. Which of the following reactions is not correctly represented?

1) $2F_2 + 2OH^- (dilute) \rightarrow 2F^- + H_2O + OF_2$

2) $2F_2 + 4OH^- (con) \rightarrow 2F^- + 2H_2O + O_2$

3) $\underset{(F_2, Cl_2, I_2)}{X_2} + \underset{(conc)}{2OH^-} \rightarrow H_2O + X^- + XO^-$

4) $Xe + F_2 \xrightarrow{light} XeF_2$

38. Which of the following interhalogen compound exists as a dimer?

1) BrF_5 2) IBr 3) ICl_3 4) ClF_3

39. Which of the following is not true of the halogens

1) all of them have seven electrons in their valency shell

2) they either gain an electron by forming an ionic bond or form a covalent compound by electron sharing

3) all exhibitvariable valency

4) all form diatomic molecules

40. Stability order of oxides formed by halogens is

1) $Cl < Br < I$ 2) $Br < Cl < I$ 3) $Br < I < Cl$ 4) $I < Br < Cl$

41. All inter halogen compounds undergo hydrolysis. ClF_5 on hydrolysis produces

1) HCl and HOF 2) HF and $HClO_3$ 3) HF and $HClO_4$ 4) HF and $HClO_2$

42. The relatively large electron-electron repulsion among the lone pairs in F_2 molecule where they are much closer to each other leads to

1) Low reactivity of F_2 molecule.

2) Smaller enthalpy of dissociation than Cl_2.

3) Smaller electron gain enthalpy than Cl_2.

4) Higher enthalpy of dissociation than Cl_2

43. The boiling points of the four hydrogen halides are given as -67^0 C, -35^0 C, $+20^0$ C and -85^0 C. What is the correct sequence of these values for HF, HCl, HBr and HI respectively?

1) $+20^o$ C, -85^o C, -67^o C, -35^o C

2) -35^0 C, -67^0 C, -85^0 C, $+20^0$ C

3) $+20^0$C, -35^0C, -85^0C, -67^0C

4) -85^0C, -67^0C, -35^0C, $+20^0$C

44. A halogen (X_2) is prepared by oxidation of its hydrogen halide, (Y). The same halogen dissolved in water on standing loses its colour due the formation of (P) and (Z). If (P) and (Y) are same, the presence of (Z) is responsible for the oxidising property of the given halogen (X_2).

Choose the incorrect statement based on the above information.

1) The catalyst used to covert X_2 to Y is $CuCl_2$.

2) Compound P gives dense white fumes with ammonium chloride.

3) The oxidation state of X in Z is +1.

4) The colour of X_2 in water is yellow.

45. The composition of bleaching powder is

1) Calcium chlorite and calcium chloride

2) Calcium chlorate and calcium hypochloride

3) Calcium hypochlorite and calcium chloride

4) Calcium chlorate and calcium hydroxide

46. The compounds X and Y are obtained by the reaction of Cl_2 with cold and dilute solution of NaOH and compounds X and Z are formed with hot and concentrated solution of NaOH. The formula of Y and Z respectively are

1) NaCl, NaClO 2) NaClO, $NaClO_3$ 3) NaCl, $NaClO_3$ 4) NaClO, HCl

47. Number of $d_\pi - p_\pi$ bonds are present in perchlorate ion and Xenon trioxide respectively are

1) 3 & 4 2) 4 & 3 3) 3 & 3 4) 2 & 3

48. How many of the following compounds are coloured?

XeF_6, $O_2[PtF_6]$, ClF_3, BrF_3, ClF, ICl_3.

1) Six 2) Three 3) Five 4) One

49. The compound in which chlorine is in its second excited state is

1) ClF_3 2) ClO_3^- 3) ClO_4^- 4) Cl_2O

50. Which of the following reactions are non -spontaneous?

1) $2KBr + Cl_2 \rightarrow 2KCl + Br_2$ 2) $2KCl + F_2 \rightarrow 2KF + I_2$

3) $2KI + Br_2 \rightarrow 2KBr + I_2$ 4) $2I_{2(s)} + 2H_2O_{(l)} \rightarrow 4I^-_{(aq)} + 4H^+_{(aq)} + O_{2(g)}$


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